

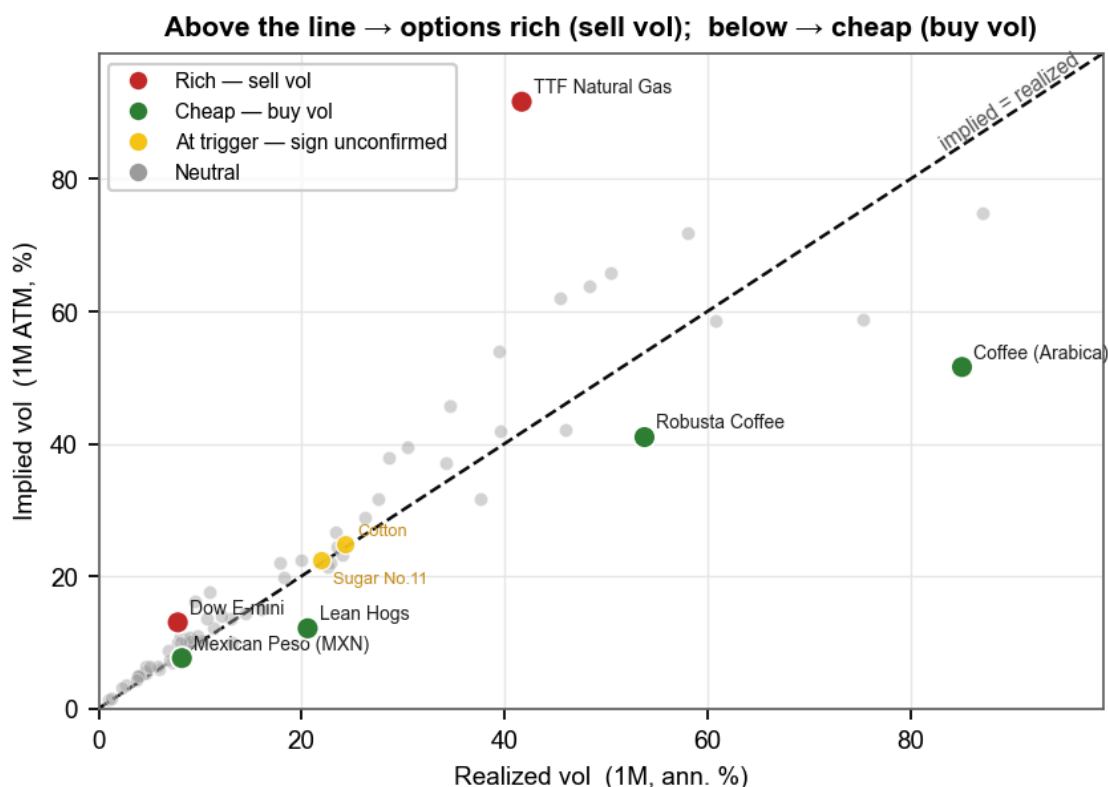
Volatility Report

Tue 21st July 2026 | XP Inc. Global Macro

Six markets stand out today. Four look potentially **cheap**, with implied sitting below the volatility the market has recently realized: **Coffee (Arabica)**, **Robusta Coffee**, **Lean Hogs** and **Mexican Peso (MXN)**. Notably, **Coffee (Arabica)** screens cheap even with its seasonal crop-weather window (☼) open — implied sitting low into a period that has historically carried genuine weather risk. Two look potentially **rich**, with implied sitting above recently realized volatility: **TTF Natural Gas** and **Dow E-mini**. One caveat (⚠): for **Coffee (Arabica)** the delivered number is fading — the last five sessions have realized only 42 against the month's 85, so part of the cheap reading reflects an event rolling out of the one-month window rather than live turbulence. They are highlighted purely as markets where the implied/realized relationship currently looks stretched, and may be worth a closer look.

How to read this. Flagged = the implied-realized spread is more than 1.5σ from its own one-year norm — high = **rich**, low = **cheap**. **Amber rows** = at the trigger but implied hasn't crossed realized (a premium compressing or a discount narrowing) — a watchlist state, not a signal. Swaps table: **shaded** = a correlated peer stretched the other way; **amber** = the 1-month correlation has drifted from the 1-year. Implieds are our own curve, built from listed option settlements — full methodology on the closing page.

IMPLIED VS REALIZED — THE SPREAD AT A GLANCE



POSSIBLE VOL SWAPS — CORRELATED $\geq 60\%$

Desk comment. The pairing that catches the eye today is **Mexican Peso (MXN)**, flagged cheap at a z of -1.6 , sitting against **South African Rand (ZAR)** at $+0.5$ — notable given the two have tracked each other at 69% over the year and 72% this past month, so that gap may be worth a closer look. Elsewhere, **Dow E-mini** and both coffees are stretched in the same direction as their closest relations, leaving no natural offset.

Flagged market (vol z)	Signal	Possible vol swaps (1y corr · 1m corr · peer's vol z)			
Dow E-mini z +1.5	Rich — sell vol	S&P 500 E-mini	76%	67%	z +0.0
		Russell 2000 E-mini	71%	47%	z +1.2
Coffee (Arabica) z -4.6	Cheap — buy vol	Robusta Coffee	69%	91%	z -2.9
Robusta Coffee z -2.9	Cheap — buy vol	Coffee (Arabica)	69%	91%	z -4.6
Mexican Peso (MXN) z -1.6	Cheap — buy vol	South African Rand (ZAR)	69%	72%	z +0.5

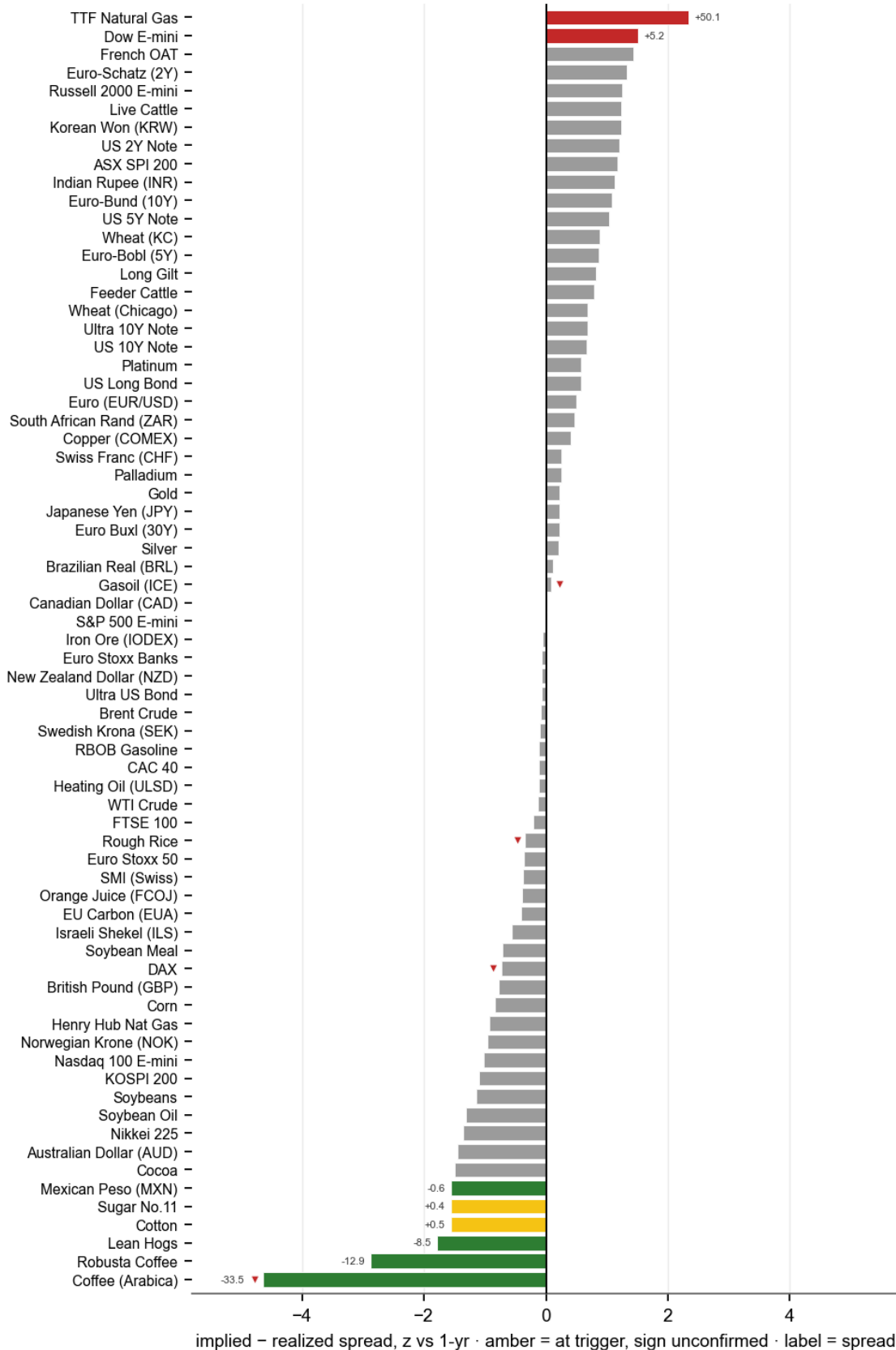
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MOST STRETCHED — RANKED BY SPREAD Z-SCORE

All markets ranked by spread z-score — flagged in colour



Red = flagged rich — the implied–realized spread is more than 1.5σ above its own one-year norm and implied sits ABOVE realized. · Green = flagged cheap — more than 1.5σ below its norm and implied sits UNDER realized. · Amber = the z is at the trigger but implied has NOT crossed realized (a premium compressing or a discount narrowing) — a watchlist state, not a signal. · Grey = inside the trigger. · ▼/▲ at a bar tip = the 5-day realized warning (see the All-Markets table): ▼ = the last five sessions realized under HALF the 1-month — the move that inflated the month has likely already happened, so read a cheap bar there with care; ▲ = more than double it — realizing is accelerating past the month.

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ALL MARKETS — FLAGGED HIGHLIGHTED

Market	Instrument	Signal	Implied 1M (±1σ)	Realized 1M (±1σ)	Realized 5d	Spread	z (1y)	%ile
TTF Natural Gas †	FJSA Comdty	Rich — sell vol ▼	91.7 (±3.448)	41.7 (±1.566)	23.0	+50.1	+2.34	100
Dow E-mini	DMA Index	Rich — sell vol ▼	13.0 (±429)	7.8 (±258)	9.6	+5.2	+1.51	94
French OAT	OATA Comdty	—	6.3 (±0.46)	4.6 (±0.34)	1.9	+1.7	+1.44	100
Euro-Schatz (2Y)	DUA Comdty	—	1.3 (±0.088)	1.0 (±0.069)	0.3	+0.3	+1.32	99
Russell 2000 E-mini	RTYA Index	—	17.6 (±33.2)	10.9 (±20.6)	13.2	+6.6	+1.25	89
Korean Won (KRW)	KOA Curncy	—	9.9 (±0.00042)	7.8 (±0.00033)	9.5	+2.2	+1.23	86
Live Cattle	LCA Comdty	—	16.1 (±2.268)	9.5 (±1.330)	15.8	+6.7	+1.23	94
US 2Y Note	TUA Comdty	—	1.6 (±0.103)	1.2 (±0.078)	1.3	+0.4	+1.20	89
ASX SPI 200	XPA Index	—	10.9 (±60)	8.0 (±44)	5.0	+2.9	+1.17	83
Indian Rupee (INR)	SIRA Curncy	—	5.4 (±0.35)	4.6 (±0.30)	1.9	+0.7	+1.13	89
Euro-Bund (10Y)	RXA Comdty	—	4.9 (±0.38)	4.0 (±0.31)	2.4	+0.9	+1.08	90
US 5Y Note	FVA Comdty	—	3.5 (±0.236)	2.7 (±0.184)	3.1	+0.8	+1.03	82
Wheat (KC) ☼	KWA Comdty	—	39.6 (±18.27)	30.4 (±14.04)	44.8	+9.2	+0.88	84
Euro-Bobl (5Y)	OEA Comdty	—	3.1 (±0.23)	2.3 (±0.16)	0.7	+0.9	+0.87	96
Long Gilt	G A Comdty	—	10.3 (±0.56)	8.8 (±0.48)	7.2	+1.5	+0.82	81
Feeder Cattle	FCA Comdty	—	13.5 (±2.981)	13.0 (±2.862)	16.9	+0.5	+0.79	72
Wheat (Chicago) ☼	W A Comdty	—	37.8 (±16.15)	28.6 (±12.23)	37.9	+9.2	+0.68	76
Ultra 10Y Note	UXYA Comdty	—	5.8 (±0.403)	4.7 (±0.330)	5.1	+1.0	+0.68	73
US 10Y Note	TYA Comdty	—	5.0 (±0.339)	4.0 (±0.273)	4.5	+1.0	+0.66	74
Platinum	PLA Comdty	—	37.0 (±38.2)	34.2 (±35.3)	22.5	+2.9	+0.58	67
US Long Bond	USA Comdty	—	8.8 (±0.615)	6.8 (±0.473)	6.9	+2.1	+0.57	72
Euro (EUR/USD)	ECA Curncy	—	5.0 (±0.00357)	3.8 (±0.00276)	3.9	+1.1	+0.50	67
South African Rand (ZAR)	RAA Curncy	—	10.0 (±0.0380)	8.6 (±0.0328)	8.8	+1.3	+0.46	70
Copper (COMEX) †	HGA Comdty	—	31.6 (±13.06)	27.5 (±11.34)	28.2	+4.2	+0.40	65
Swiss Franc (CHF)	SFA Curncy	—	6.3 (±0.488)	5.8 (±0.450)	7.1	+0.5	+0.26	59
Palladium	PAA Comdty	—	41.9 (±33.9)	39.6 (±32.1)	24.2	+2.3	+0.25	54
Japanese Yen (JPY)	JYA Curncy	—	6.3 (±0.243)	5.0 (±0.194)	3.2	+1.4	+0.22	59
Gold	GCA Comdty	—	22.1 (±56.8)	22.8 (±58.5)	17.9	-0.7	+0.22	45
Euro Buxl (30Y)	UBA Comdty	—	10.8 (±0.72)	8.9 (±0.60)	8.6	+1.9	+0.22	53
Silver	SIA Comdty	—	42.1 (±1.567)	46.0 (±1.714)	41.3	-4.0	+0.20	44
Brazilian Real (BRL)	BRA Curncy	—	10.2 (±0.127)	9.0 (±0.111)	6.1	+1.2	+0.12	51
Gasoil (ICE) †	QSA Comdty	—	71.8 (±54.53)	58.1 (±44.14)	21.1 ▼ 36%	+13.7	+0.09	56
Canadian Dollar (CAD)	CDA Curncy	—	4.3 (±0.193)	3.7 (±0.165)	4.4	+0.6	+0.03	44
S&P 500 E-mini	ESA Index	—	13.5 (±64.12)	10.6 (±50.52)	11.5	+2.9	+0.02	47
Iron Ore (IODEX) †	SCOA Comdty	—	9.9 (±0.62)	13.2 (±0.82)	9.7	-3.3	-0.06	49
Ultra US Bond	WNA Comdty	—	9.9 (±0.697)	8.1 (±0.570)	8.0	+1.8	-0.07	49
New Zealand Dollar (NZD)	NVA Curncy	—	7.3 (±0.269)	7.0 (±0.259)	6.0	+0.3	-0.07	49
Euro Stoxx Banks	SX7E Index	—	22.4 (±4.26)	20.0 (±3.82)	19.2	+2.3	-0.07	50
Brent Crude †	COA Comdty	—	63.8 (±3.65)	48.4 (±2.77)	32.0	+15.4	-0.08	67
Swedish Krona (SEK)	SEA Curncy	—	8.0 (±0.0518)	7.6 (±0.0497)	8.0	+0.3	-0.10	45
RBOB Gasoline	XBA Comdty	—	54.0 (±11.58)	39.5 (±8.47)	25.4	+14.1	-0.11	55
Heating Oil (ULSD)	HOA Comdty	—	65.9 (±17.12)	50.5 (±13.12)	22.4	+14.9	-0.12	56
CAC 40	CAC Index	—	14.0 (±73.87)	12.1 (±63.57)	4.6	+2.0	-0.12	44
WTI Crude †	CLA Comdty	—	62.1 (±3.30)	45.5 (±2.42)	32.4	+16.6	-0.13	69
FTSE 100	UKX Index	—	11.0 (±73.26)	9.8 (±65.39)	8.6	+1.2	-0.21	33
Rough Rice ☼	RRA Comdty	—	21.4 (±0.191)	22.6 (±0.202)	11.1 ▼ 49%	-1.2	-0.34	43
Euro Stoxx 50	SX5E Index	—	14.4 (±57.18)	14.4 (±56.94)	10.4	+0.1	-0.36	32
SMI (Swiss)	SMI Index	—	12.2 (±110.27)	11.3 (±101.64)	8.1	+1.0	-0.37	39
Orange Juice (FCOJ)	JOA Comdty	—	58.5 (±5.30)	60.8 (±5.51)	67.2	-2.3	-0.39	33
EU Carbon (EUA)	MOA Comdty	—	28.8 (±1.51)	26.3 (±1.38)	44.5	+2.2	-0.40	28
Israeli Shekel (ILS)	ISA Curncy	—	10.1 (±0.00208)	10.1 (±0.00209)	10.5	+0.0	-0.56	29
Soybean Meal ☼	SMA Comdty	—	22.1 (±4.6)	17.8 (±3.7)	17.6	+4.2	-0.71	24
DAX	DAX Index	—	15.0 (±236.47)	16.0 (±252.62)	7.8 ▼ 49%	-1.0	-0.73	17
British Pound (GBP)	BPA Curncy	—	5.9 (±0.49)	6.0 (±0.51)	10.5	-0.1	-0.77	21
Corn ☼	C A Comdty	—	26.7 (±7.98)	23.3 (±6.97)	18.3	+3.4	-0.84	24
Henry Hub Nat Gas	NGA Comdty	—	45.7 (±0.082)	34.6 (±0.062)	27.3	+11.1	-0.93	23
Norwegian Krone (NOK)	NOA Curncy	—	7.8 (±0.0512)	8.3 (±0.0539)	5.5	-0.4	-0.96	21
Nasdaq 100 E-mini	NQA Index	—	23.2 (±428.07)	24.0 (±443.90)	22.4	-0.9	-1.02	16
KOSPI 200	KOSPI2 Index	—	74.9 (±50.63)	87.1 (±58.88)	90.8	-12.2	-1.10	12
Soybeans ☼	S A Comdty	—	19.8 (±15.28)	18.2 (±14.03)	15.7	+1.6	-1.14	13
Soybean Oil ☼	BOA Comdty	—	24.5 (±1.11)	23.5 (±1.06)	21.7	+1.0	-1.31	5
Nikkei 225	NKY Index	—	31.7 (±1322.34)	37.6 (±1567.81)	47.9	-5.9	-1.36	10
Australian Dollar (AUD)	ADA Curncy	—	6.9 (±0.303)	7.2 (±0.318)	4.6	-0.3	-1.45	7
Cocoa	CCA Comdty	—	58.8 (±208)	75.4 (±266)	78.4	-16.5	-1.49	8
Cotton ☼	CTA Comdty	Premium compressing	24.9 (±1.26)	24.3 (±1.23)	28.5	+0.5	-1.56	10
Mexican Peso (MXN)	PEA Curncy	Cheap — buy vol ▲	7.6 (±0.027)	8.2 (±0.029)	8.0	-0.6	-1.56	6
Sugar No.11	SBA Comdty	Premium compressing	22.4 (±0.21)	21.9 (±0.21)	30.9	+0.4	-1.56	9
Lean Hogs	LHA Comdty	Cheap — buy vol ▲	12.1 (±0.669)	20.6 (±1.141)	18.6	-8.5	-1.79	2
Robusta Coffee	DFA Comdty	Cheap — buy vol ▲	41.0 (±99)	53.8 (±130)	34.1	-12.9	-2.87	0
Coffee (Arabica) ☼ †	KCA Comdty	Cheap — buy vol ▲	51.6 (±10.48)	85.1 (±17.28)	41.7 ▼ 49%	-33.5	-4.64	1

☼ seasonal weather windows currently open — Wheat (KC): Apr–Jul wheat weather · Wheat (Chicago): Apr–Jul wheat weather · Rough Rice: May–Aug growing season · Soybean Meal: Jun–Sep soy weather · Corn: Jun–Aug US weather/pollination · Soybeans: Jun–Sep US weather · Soybean Oil: Jun–Sep soy weather · Cotton: Jun–Aug US cotton weather · Coffee (Arabica): Jun–Aug Brazil frost risk

† our curve prints 3+ vols away from the vendor's 30-day surface — TTF Natural Gas +8.8 · Copper (COMEX) +3.5 · Gasoil (ICE) +5.2 · Iron Ore (IODEX) -7.0 · Brent Crude +3.9 · WTI Crude +11.1 · Coffee (Arabica) +3.6; the table shows our number, taken from listed settlements. Realized 5d = the same calculation as Realized 1M over only the last five sessions. ▼ n% = 5d under half the 1-month (shaded share shown); the move that inflated the month has likely already happened, so a cheap reading there partly reflects the past. ▲ = 5d more than double the 1-month (realizing accelerating). Both need two consecutive sessions.

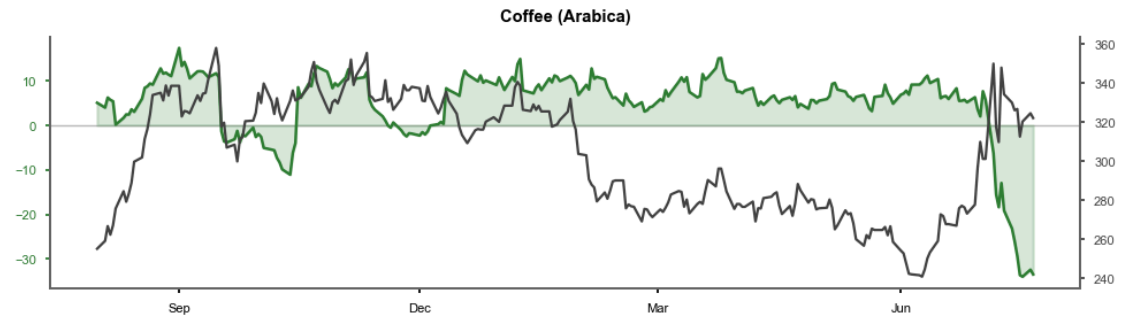
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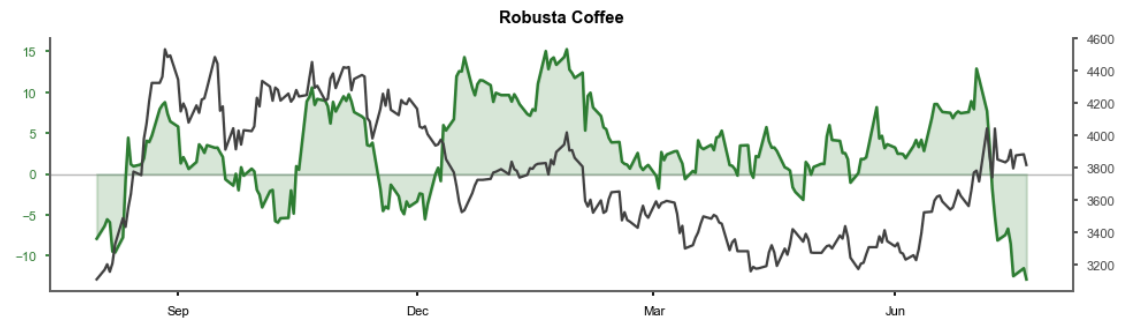
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FLAGGED — SPREAD VS UNDERLYING (1 YEAR)

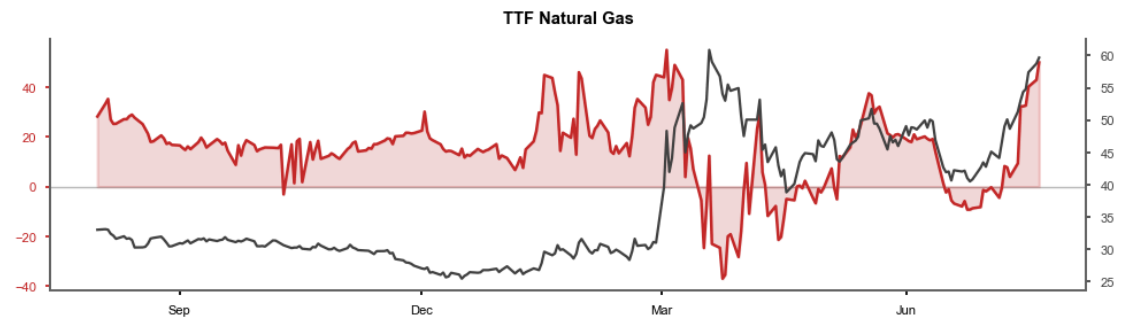
shaded / coloured: implied – realized (vol pts, left axis) · grey: underlying price (right axis)



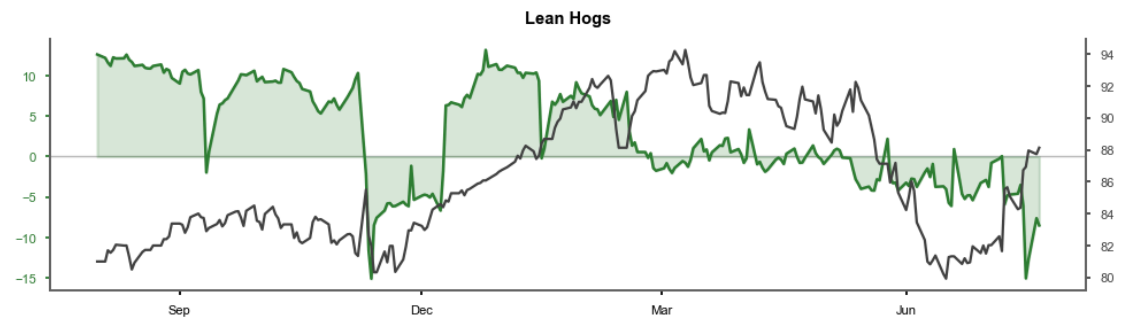
The spread sits at **-33.5** vol points — implied of **51.6** against **85.1** realized — its most depressed reading of the past year at the **1st** percentile, after a month in which implied added **+13.5** points while realized surged **+55.3** and the underlying gained **+21%**. The chart also falls within the seasonal weather window (Jun–Aug Brazil frost risk), when elevated implied partly prices genuine crop risk.



At the **0th** percentile of the past year, this is the most depressed the spread has been over that stretch: implied of **41.0** against **53.8** realized leaves a gap of **-12.9** vol points, with implied up **+7.5** points over the past month versus **+27.9** for realized as the underlying climbed **+8%**.



Implied at **91.7** now towers **+50.1** vol points above realized at **41.7**, a **100th** percentile reading last seen in **early March**, when the underlying had moved **+63%** in a month and realized was running **113**. This time the gap has opened differently, with implied up **+38.4** points over the past month while realized fell **-19.6** and the underlying rose **+42%**.



With implied at **12.1** against **20.6** realized, the spread of **-8.5** vol points sits at the **2nd** percentile of the past year, a level last touched in **mid November 2025** when realized was running **23**. Over the past month implied has slipped **-1.0** points while realized edged up **+2.9**, with the underlying **+9%** over the same stretch.

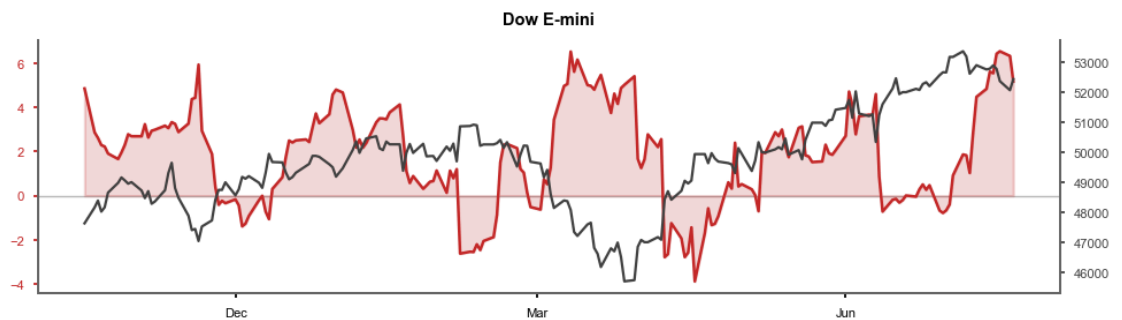
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A narrow gap here: implied of **7.6** versus **8.2** realized puts the spread at **-0.6** vol points, the **6th** percentile of the past year and last seen this depressed in **late April**, when the underlying had moved **+3%** in a month and realized was running **10**. Over the past month implied has drifted **-0.6** points while realized picked up **+1.6**.



The spread has widened to **+5.2** vol points — implied of **13.0** over **7.8** realized — placing it at the **94th** percentile, a reading last seen in **late March** when the underlying had moved **-8%** in a month with realized running **14**. Notably, the widening this time has come from realized falling **-6.6** points over the past month against a milder **-1.4** decline in implied.

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SHORT-TERM RATES — RATE VOL

The headline figures are the 1-day 1σ move in basis points — the number the desk actually trades (percentage vols, grey, vary with each rate's level and aren't comparable across currencies). Implied is **our own constant 90-day ATM**, built identically for all three markets — listed quarterly option settlements inverted through Black-76 on the **implied rate** ($100 - \text{price}$), ATM call/put mids interpolated in total variance — with Bloomberg's published 3-month point shown as vs *BBG* where it exists (none is published for Euribor). Realized is the matching 63-day delivered vol of the implied rate; implied naturally sits above it — options price the meetings ahead — so the flag is each market's **z against its own year**, and a directional flag also requires the spread's sign to agree: a stretched z on a still-negative spread shows as *Discount narrowing*, not a sale. 1M SOFR and Fed Funds options are too thin for a reliable mark; €STR has no listed options.

3M SONIA (Short Sterling) SFIA Comdty

Rich — sell vol ▼

RATE 3.76% · IMPLIED **±4.3bp** (18.2 vol) · REALIZED **±1.8bp** (7.7 vol) · SPREAD +10.4 · Z (1Y) +3.28 · %ILE 100 · VS BBG -2.7

3M SOFR SFRA Comdty

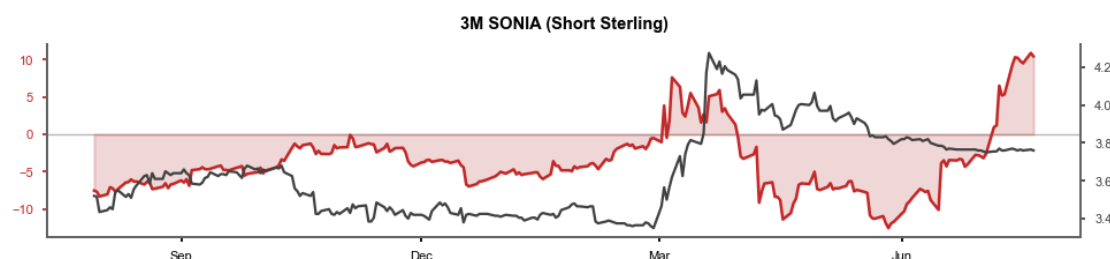
Discount narrowing

RATE 4.08% · IMPLIED **±4.5bp** (17.3 vol) · REALIZED **±5.1bp** (19.7 vol) · SPREAD -2.4 · Z (1Y) +1.54 · %ILE 98 · VS BBG -1.9

3M Euribor ERA Comdty

RATE 2.79% · IMPLIED **±3.3bp** (18.6 vol) · REALIZED **±5.3bp** (30.3 vol) · SPREAD -11.4 · Z (1Y) +0.23 · %ILE 43 · VS BBG n/a

flagged — 1 year: shaded / coloured = implied – realized (rate-vol points, left axis) · grey = implied rate % (right axis)



DEFINITIONS & METHODOLOGY

Implied vs realized — what options price for the next 30 days against what the market delivered (21-day close-to-close, annualised). The implied is **our own** constant-30-day curve: ATM call and put settlements of every listed expiry, fitted per product (FX: OTC pair vol; equity cash indices: vendor surface). Implied looks forward, realized back — a positive spread isn't automatically mispricing.

($\pm 1\sigma$) — the 1-day one-standard-deviation move in the contract's own units ($\text{vol} \div \sqrt{252} \times \text{price}$).

Possible vol swaps — closest peers by 1-year realized-vol correlation (1y · 1m; amber = the 1-month has drifted; shaded = a peer stretched the other way).

Short-term rates — our own constant-90-day construction from listed option settlements (see the section note): both legs are rate vols, the moves in basis points, each judged against its own year.

Sources & timing. Implied vols are **our own construction** for 44 of the 70 markets: the ATM call and put settlements of every listed option expiry, inverted through Black-76 and fitted to a constant-30-day curve per product (the short-term-rate section: the same method at 90 days). FX implieds are OTC pair vols and equity cash indices use the vendor's listed-option surface; the surface also backstops any market our build cannot price on the day. † marks markets where our curve and the vendor's 30-day surface disagree by 3+ vols. Realized vols are our own calculation from settlement prices. Figures are computed from the **Tue 21st July 2026** settlements, pulled 22 Jul 2026, 06:20. Generated by the internal Strategy Monitor for information only — not investment advice or a solicitation; source data is subject to your market-data licence terms.

Spread · z · %ile — spread = implied – realized (vol pts); the z-score sets it against the market's own trailing year ($|z| \geq 1.5$ flags); the percentile ranks it (100 = widest of the year).

1M · 5d · ▼ · ▲ — the 1M pair is what the spread compares (implied 30 days forward, realized 21 sessions back); 5d = the last five sessions only. ▼ n% = 5d under half the month — the inflating move likely already happened; ▲ = more than double it. Two consecutive sessions required.

☀ **seasonal window** — an ag market in its crop-weather months; a rich reading there partly prices genuine weather risk.

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